JAVES-AI ARCHITECTURE

EFFECTIVE 27 OCT 2026

Current decisions and ownership boundaries

This page supersedes any conflicting label on the original diagrams that follow.

1 External CCTV Archive

The existing camera archive is read-only and outside JAVES-AI. Only Perception and Clip resolve archive_id. Main, Routing and Retention never call or delete it.

2 Private filesystem storage

No separate cloud storage is planned. Product files live under JAVES_ARTIFACT_ROOT through FileArtifactStore; object_key is a safe relative path.

3 Fourteen-day case retention

Search-only data is removed on close. Final route, confirmed clips and export are physically removed 14 calendar days after close. Legal hold prevents deletion.

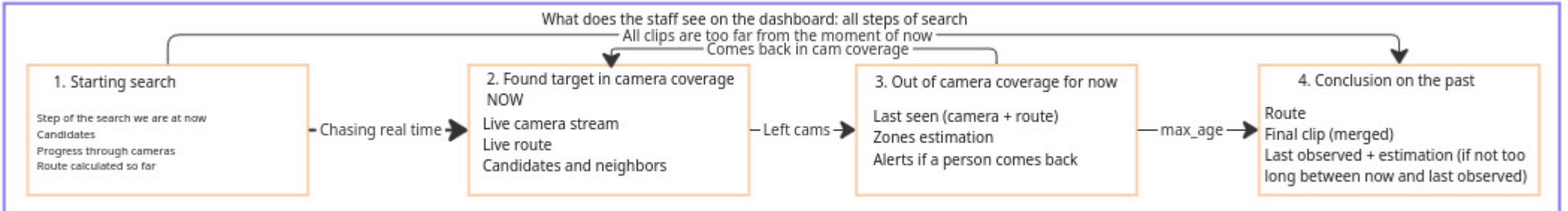
4 Permanent security audit

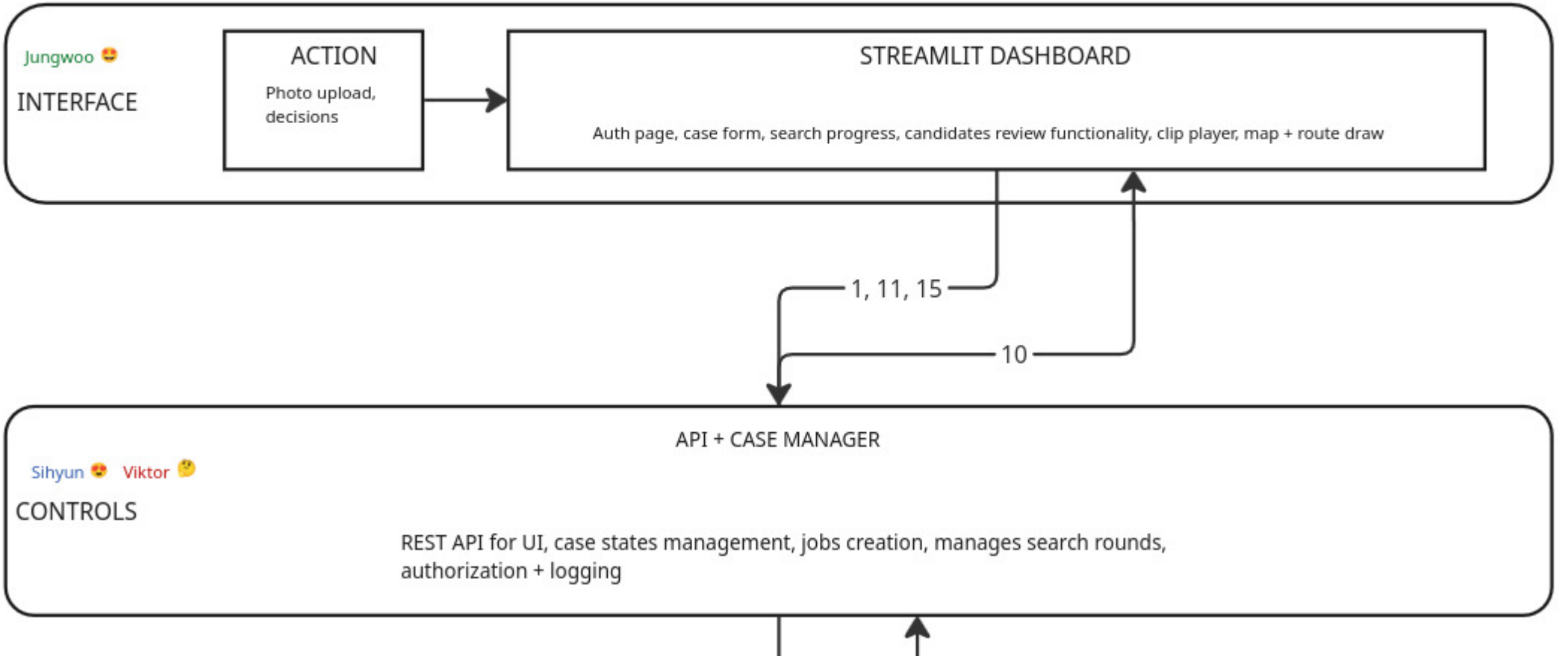
Actor, timestamp, action and details are append-only and retained forever. Case DELETE and TTL never redact or remove audit events.

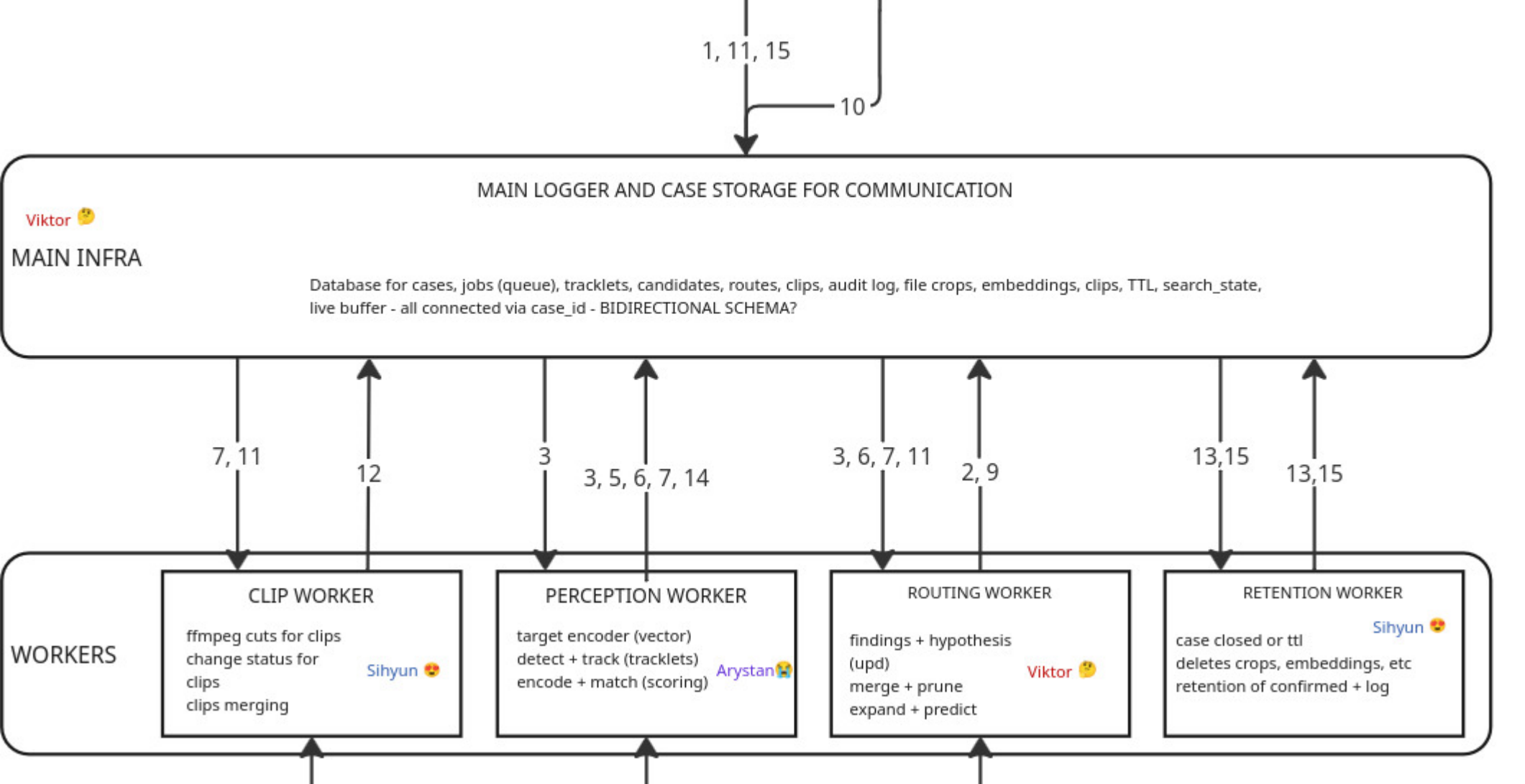
5 Routing owns route rendering

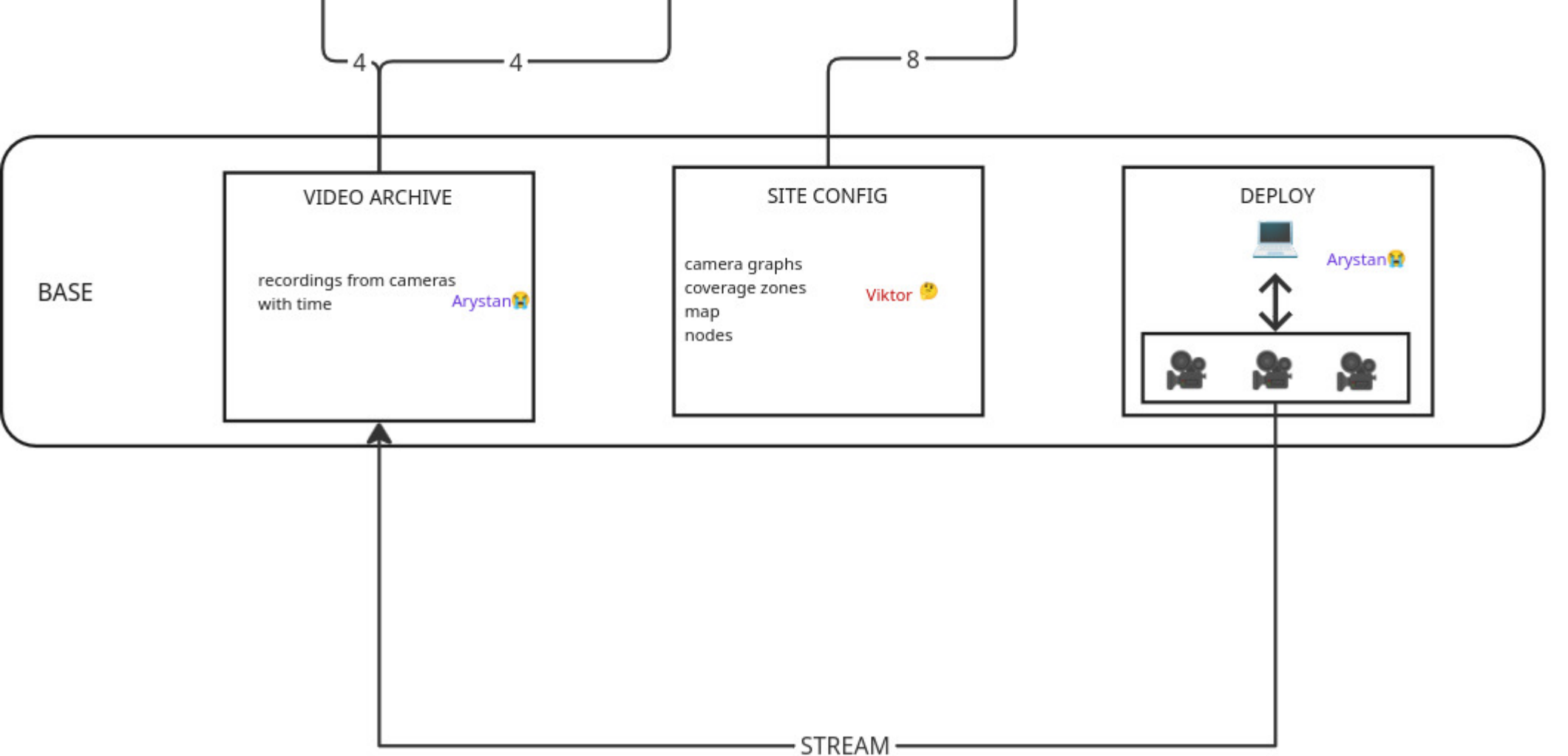
Routing calculates the route and returns ready per-floor SVG maps. Dashboard displays those SVGs and never redraws route geometry.

Implemented details: docs/architecture.md | Documentation map: docs/README.md


What does the staff see on the dashboard: all steps of search
All clips are too far from the moment of now
Comes back in cam coverage
1. Starting search
Step of the search we are at now
Candidates
Progress through cameras
Route calculated so far
Chasing real time
2. Found target in camera coverage NOW
Live camera stream
Live route
Candidates and neighbors
Left cams
3. Out of camera coverage for now
Last seen (camera + route)
Zones estimation
Alerts if a person comes back
max_age
4. Conclusion on the past
Route
Final clip (merged)
Last observed + estimation (if not too long between now and last observed)

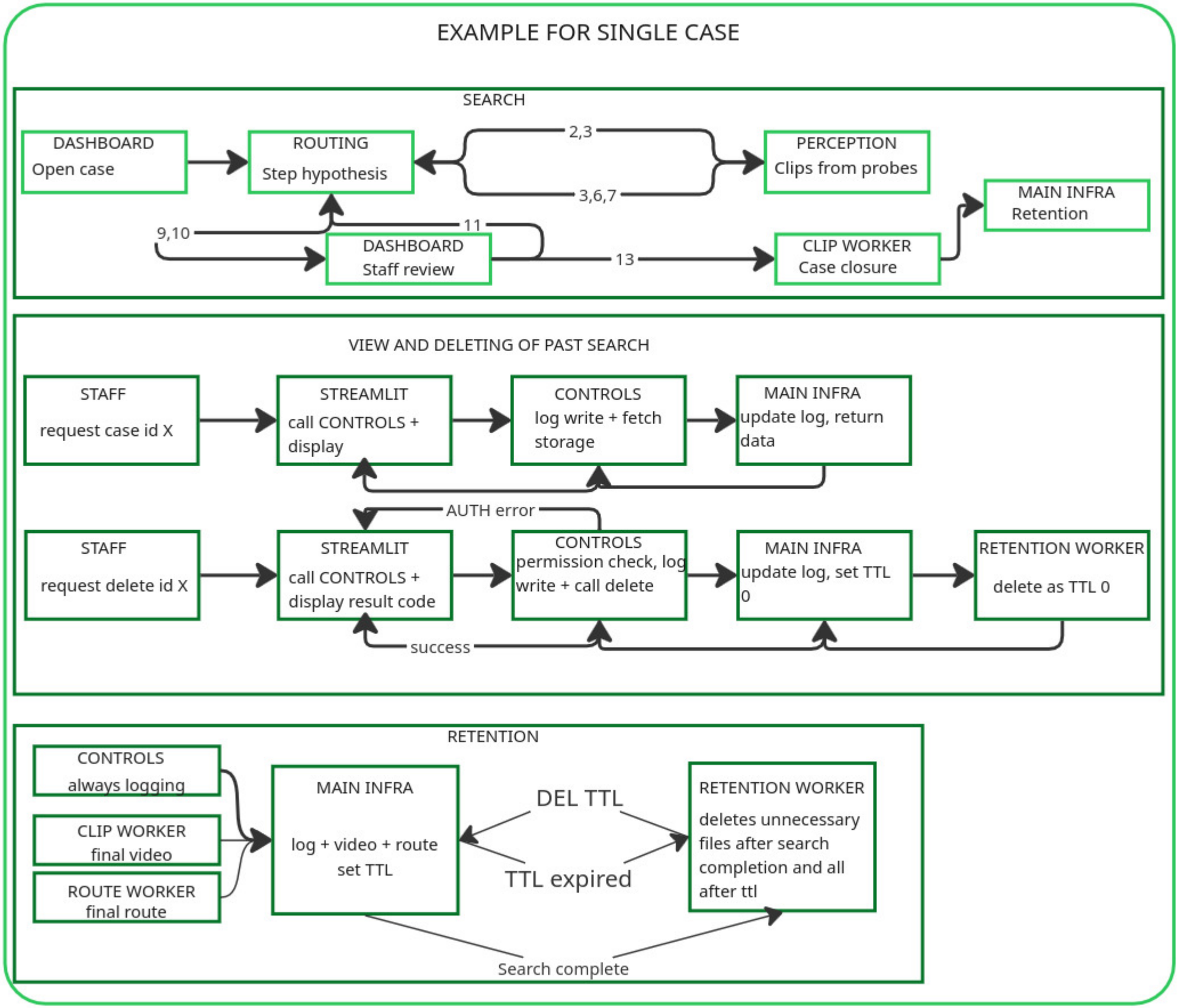

Jungwoo
INTERFACE
ACTION
Photo upload, decisions
STREAMLIT DASHBOARD
Auth page, case form, search progress, candidates review functionality, clip player, map + route draw
1, 11, 15
10
API + CASE MANAGER
Sihyun
Viktor
CONTROLS
REST API for UI, case states management, jobs creation, manages search rounds, authorization + logging


1, 11, 15
10
MAIN LOGGER AND CASE STORAGE FOR COMMUNICATION
Viktor
MAIN INFRA
Database for cases, jobs (queue), tracklets, candidates, routes, clips, audit log, file crops, embeddings, clips, TTL, search_state, live buffer - all connected via case_id - BIDIRECTIONAL SCHEMA?
7, 11
12
3
3, 5, 6, 7, 14
3, 6, 7, 11
2, 9
13,15
13,15
WORKERS
CLIP WORKER
ffmpeg cuts for clips
change status for clips
clips merging
Sihyun
PERCEPTION WORKER
target encoder (vector)
detect + track (tracklets)
encode + match (scoring)
Arystan
ROUTING WORKER
findings + hypothesis (upd)
merge + prune
expand + predict
Viktor
RETENTION WORKER
Sihyun
case closed or ttl
deletes crops, embeddings, etc
retention of confirmed + log


4
4
8
BASE
VIDEO ARCHIVE
recordings from cameras
with time
Arystan😭
SITE CONFIG
camera graphs
coverage zones
map
nodes
Viktor 🤔
DEPLOY
Arystan😭
STREAM

COMMUNICATION

1. Case request (reference image, last known {camera | zone, time}, window {start, end}, operator_id fields) - when creating case
2. Probe request ({probes[{probe_id, camera_id, t_from, t_to | OPEN, priority, reason, hypothesis_ids[]}], mode, done, stop_reason} fields) - probes for each camera step t_to = OPEN is live cam subscription
3. Probe job ({job_id, probe_id, kind: TARGET | PROBE, camera_id, t_from, t_to | OPEN} fields) - answer like status: DONE | FAILED | TIMEOUT, result {n_tracklets, n_above_floor, best_score}
4. Frames (get_frames(camera_id, t_from, t_to) -> frames, absolute time ; and get_segments(...) gives file for clip now() to get source time)
5. Target embedding ({case_id, vector[512], source: reference | confirmed_tracklet, crop_path, quality} fields) - is a gallery inside each case (tracklets added after staff confirmation)
6. Tracklet ({tracklet_id, camera_id, t_start, t_end, open, clipped_start, clipped_end, bbox_first, bbox_last, embedding, crop_paths[], clip_ref} fields) - open means track ongoing
7. Candidate ({candidate_id, tracklet_id, appearance_score $\in$ [0, 1], cosine}) - routing adds reject reason and final score
8. Site graph (cameras[], regions[], edges[] {id, t_min, t_typ, t_max, d_m}, blind_nodes[] {dwell}, outside {entrances{camera: p}, dwell}, map, clock_offset_s) for yaml config
9. Hypothesis + estimate (hypotheses[] {id, visits[], gap, score, status: ALIVE | DORMANT | ENDED}, best, ambiguities[], estimate {zones[{zone, p}], valid_until}, mode) - search_state read only by Routing worker
10. Case view - All the dashboard needs: mode, search progress, candidates (thumbnail, score, status), best hypothesis & alternatives, estimate, live {camera_id, stream_url, bbox, track_age_s}, watch[], clips. UI reads only this. 2 updates per second in search mode.
11. Review desicion (candidate_id, decision: confirm | reject, operator_id (staff), timestamp fields) - to change clip state and recalculate route fields)
12. Clip record ({clip_id, candidate_id, camera_id, t_from, t_to, state: HOLD | CONFIRMED | RELEASED, file}. export_merged(case_id)) - merges clips with export_merged(case_id)
13. Case closure ({case_id, closed_by}) - delete all except final route, confirmed merged clip + log
14.Live feed - feed with drawn target box in live-buffer (files/live/{case_id}/{camera_id}/, TTL secs) and line live_state {camera_id, track_id, bbox, age_s}. API gives buffer as MJPEG or HLS; in CaseView that is stream_url.
15.State change (SAVE | DELETE case, case_id, operator_id, timestamp)


EXAMPLE FOR SINGLE CASE
SEARCH
DASHBOARD
Open case
ROUTING
Step hypothesis
2,3
3,6,7
PERCEPTION
Clips from probes
MAIN INFRA
Retention
9,10
11
DASHBOARD
Staff review
13
CLIP WORKER
Case closure
VIEW AND DELETING OF PAST SEARCH
STAFF
request case id X
STREAMLIT
call CONTROLS + display
CONTROLS
log write + fetch storage
MAIN INFRA
update log, return data
AUTH error
STAFF
request delete id X
STREAMLIT
call CONTROLS + display result code
CONTROLS
permission check, log write + call delete
MAIN INFRA
update log, set TTL 0
RETENTION WORKER
delete as TTL 0
success
RETENTION
CONTROLS
always logging
CLIP WORKER
final video
ROUTE WORKER
final route
MAIN INFRA
log + video + route
set TTL
DEL TTL
TTL expired
RETENTION WORKER
deletes unnecessary files after search completion and all after ttl
Search complete